MART-NILM: A Compact Multi-Appliance Relational Temporal Network for Joint Power and State Estimation in Non-Intrusive Load Monitoring

Raymond Tie Siou Hin, Tan Wen Shan, Ding Ze Yang, Mirza Rayana Sanzana, and Haval Kamil

***Abstract*—Appliance-level load visibility supports residential demand-side management, but aggregate meters do not directly reveal individual consumption or operating states. Non-intrusive load monitoring (NILM) estimates these quantities without dedicated appliance sensors. Multi-appliance disaggregation shares computation across loads, yet sparse activations under limited background conditions and unequal power–state loss scales complicate joint learning. This paper proposes the Multi-Appliance Relational Temporal Network for NILM (MART-NILM), a compact joint estimator. A shared temporal convolutional encoder, appliance-specific decoders, and relational attention jointly estimate five appliance powers and states. Focal-event background-stratified mixing exposes active appliance traces to varied residual loads, while bounded appliance-wise balancing adjusts each classification objective relative to its corresponding regression loss. Both strategies operate only during training. Experiments on two public residential datasets use disjoint training, validation, and test houses and preserve recorded aggregates. Across three seeds, MART-NILM achieves appliance-average mean absolute errors of 7.5 and 10.8 watts and macro-averaged F1 scores of 0.88 and 0.72 on the respective test houses, outperforming three reproduced baselines on both measures. With 1.373 million parameters and lower training time under the evaluated configurations, the method offers a favorable estimation–cost trade-off for residential load monitoring.**

***Index Terms*—non-intrusive load monitoring, energy disaggregation, multi-task learning, power and state prediction, household appliances**

I. Introduction

Residential energy management requires information about both the magnitude and operating state of individual loads. Aggregate metering provides household demand, but not this appliance-level visibility. Dedicated submetering can supply detailed feedback [1], at the cost of installing and maintaining additional sensors. Non-intrusive load monitoring (NILM) estimates appliance consumption from aggregate measurements and offers a less instrument-intensive basis for residential load analysis [2], [3]. To support residential load observability, a NILM estimator must identify appliance activity as well as reconstruct appliance power: low average power error can conceal missed operating periods, particularly for appliances that remain off for most of the record.

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Classical NILM identifies appliance signatures and infers appliance states with finite-state or factorial hidden Markov models [4], [5]. Deep approaches instead learn representations directly from aggregate windows [6], reducing the need to specify signatures and state-transition models manually. Regardless of the estimator, a recorded aggregate contains demand outside the monitored target set. For N target appliances, the measured power is

$$x_t = \sum_{i=1}^{N} P_{t,i} + b_t + \epsilon_t, \tag{1}$$

where $P_{t,i}$ is target power, b_t is unmonitored household demand, and ϵ_t represents measurement and timing discrepancies. This residual background can exceed a low-power target and varies between households, so a familiar appliance event can occur in an unfamiliar aggregate context. House-held-out evaluation on recorded aggregates is therefore needed; results from different sensing and partitioning protocols are not directly interchangeable [7].

Within this setting, two common deep NILM modelling paradigms are single-appliance and multi-appliance estimation. Single-appliance networks predict one target per model and have advanced disaggregation accuracy [8]–[10], but repeating a model for every load repeats aggregate feature extraction and increases storage requirements. Recent deep NILM work has therefore increasingly studied multi-appliance estimation, in which one model jointly estimates a fixed set of N target appliances from the same aggregate window [11]–[14]. NILM research commonly considers two complementary tasks: power estimation and ON–OFF state detection. For appliance i at time t, these tasks estimate a power trace $\hat{y}_{t,i}$ and an ON-state probability $\hat{s}_{t,i}$, respectively. Power estimation is a regression task, whereas state detection is multi-label classification in the multi-appliance setting because several appliances may be ON at the same time. For an aggregate window $\mathbf{x}$ of length T, the joint estimator is

$$\left(\hat{\mathbf{Y}}, \hat{\mathbf{S}}\right) = f_{\boldsymbol{\theta}}(\mathbf{x}), \qquad \hat{\mathbf{Y}} \in \mathbb{R}^{T\times N}, \quad \hat{\mathbf{S}} \in [0,1]^{T\times N}, \tag{2}$$

where $\hat{\mathbf{Y}}$ and $\hat{\mathbf{S}}$ collect the estimated power traces and ON-state probabilities, respectively.

Joint multi-appliance and multi-label estimation is more challenging than running N independent predictors. Concurrent activations provide useful cross-appliance context, but their superposed transitions can mask a weak target beneath other loads or the residual background [11], [12], [15]. A

joint architecture must distinguish appliance-specific evidence within the shared aggregate and support coherent power–state outputs. Explicit cross-appliance interaction offers one way to incorporate relational context. Shared feature extraction can also reduce repeated computation across appliances.

Training raises two further difficulties. First, sparse events in a finite record occur under only a subset of residual-background conditions. A model can then use residual-background correlations as a shortcut instead of the appliance signature; this can fail when those correlations change across households [16]. Second, appliance power ranges and on-state prevalences differ substantially. Existing joint objectives use a shared regression–classification balance, either as a fixed coefficient or a global loss-ratio weight [14], [17]. Such a shared balance cannot represent appliance-specific scales; when derived from joint losses, a large power error for one appliance also changes the relative state weight of the others. These observations motivate control over both residual-background conditions and the power–state balance of each appliance.

To address these issues, we propose the Multi-Appliance Relational Temporal Network for NILM (MART-NILM), a relational temporal model for joint power and ON–OFF state estimation under residual-background variation.

The main contributions are threefold:

1) A compact multi-appliance architecture, MART-NILM, is designed for joint power and state estimation. It extracts a shared temporal representation once and combines appliance-specific decoding with cross-appliance relational context to produce five paired power–state outputs in a single forward pass.
2) Two training-only strategies address sparse events, residual-background variation, and appliance-wise power–state imbalance. Focal-event background-stratified mixing exposes activations to a controlled range of recorded residual-background conditions, while bounded appliance-wise balancing independently aligns the power and state objectives of each appliance. Neither strategy adds inference-time computation.
3) A house-held-out evaluation uses recorded aggregates from UK-DALE and REFIT. Three-seed baseline comparisons, architecture and training ablations, and background-conditioned analyses quantify power estimation, state detection, computational cost, and remaining failure modes.

II. Related Work

Multi-appliance and multi-label NILM estimates power and predicts an ON–OFF label for each appliance in a fixed set from a single aggregate measurement. This setting has a nested multi-task structure: the inner level couples power regression and state classification for each appliance, whereas the outer level determines how representations are shared across appliances that may be active concurrently. Existing NILM studies recognize these two levels through formulations described as "multi-task-multi-appliance" [11] and "multi-appliance-task" [14], [15].

Most generic multi-task learning (MTL) formulations regard tasks as a single peer set and learn which parameters, experts, or intermediate representations should be shared across them [18], [19]. In multi-appliance and multi-label NILM, however, task relations are structured rather than flat. The model must couple heterogeneous power-regression and state-classification targets within each appliance while separating and relating appliances that may be active concurrently in the same aggregate. A suitable sharing design must therefore preserve appliance-specific signatures without discarding useful cross-appliance context.

Relevant NILM studies, including multi-appliance and multi-label formulations, use convolutional neural networks (CNNs) [11], [12], recurrent neural networks (RNNs) [20], temporal convolutional networks (TCNs) with attention [15], [21], and Transformers [14]. Backbone choice controls temporal feature extraction; the remaining architectural question is how those features are organized for joint prediction. In joint multi-appliance and multi-label estimation, this organization entails two linked design decisions: *appliance-head design*, which governs within-appliance power–state output coupling, and *cross-appliance body design*, which governs where appliance representations separate, remain shared, or exchange information before reaching the appliance heads. Figures 1 and 2 summarize these two levels, respectively.

At the appliance-head level, early joint NILM designs use separate state and power branches, with the predicted state concatenated with power-branch features before the final regression layers. The CNN and RNN models of Murray *et al.* [20], illustrated in Fig. 1(a), use this state-informed concatenation. It allows the regressor to use state information while retaining direct power features, but does not explicitly enforce consistency between OFF predictions and power estimates.

An explicit connection is introduced by the subtask-gated network (SGN) [22], which multiplies the regression output by a separately supervised on-state probability, as shown in Fig. 1(b). This gate suppresses power during predicted inactivity, although a missed activation also attenuates an otherwise valid power estimate. The basic formulation suppresses standby consumption, motivating a standby-power variant in the original study [22].

To accommodate different event durations and temporal contexts, the scale- and context-aware network (SCANet) [23] extends SGN with parallel dilated branches and self-attention. Figure 1(c) summarizes this design: classification features gate the corresponding regression features at each scale, followed by separate feature fusion and contextual processing, while the final output retains state gating. This richer coupling supports feature selection at multiple temporal scales, at the cost of additional branches and attention computation, and retains sensitivity to state-derived gates. Thus, Fig. 1(c) extends the gating design in Fig. 1(b), rather than representing an unrelated alternative. These appliance-head mechanisms can be combined with different ways of sharing information across appliances.

The resulting design question is where to share information across appliances. Multi-task architectures commonly share an

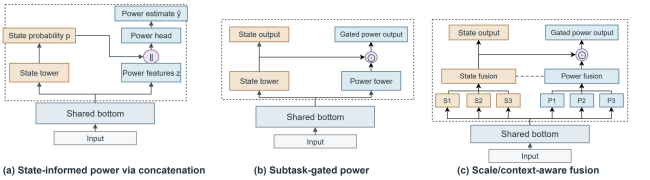


Fig. 1. Within-appliance power–state heads: (a) concatenation [20], (b) output gating [22], and (c) multi-scale fusion [23].

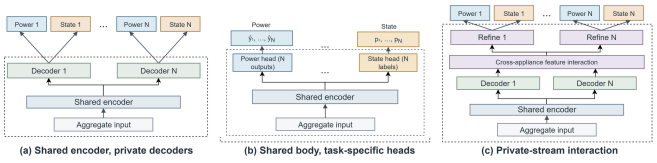


Fig. 2. Multi-appliance body designs: (a) shared encoder/private decoders [15], (b) shared body/task heads [11], and (c) private-stream interaction.

encoder and retain task-specific outputs, or continue exchanging information during decoding [18]. MATNilm explicitly branches by appliance and then by regression and classification task [15]; Fig. 2(a) isolates this hierarchical branch pattern. Its full decoder also includes temporal and appliance-wise attention. Shared encoding reduces repeated feature extraction, while private branches retain capacity for different appliance signatures.

A fully shared body with task-specific output heads is another joint-estimation design. As shown in Fig. 2(b), it retains one representation for all appliances, then uses separate heads to predict the vector of appliance powers and the vector of independent appliance-state labels. UNet-NILM is one example of this pattern [11]. This approach shares all body features, rather than maintaining appliance-specific streams before prediction.

Feature interaction, illustrated in Fig. 2(c), provides a complementary mechanism. In dense prediction, attention-guided message passing transfers information from intermediate task predictions to final tasks [24]. In NILM, MATNilm already models dependencies across appliances and time [15]. MART-NILM applies this information-sharing principle directly to appliance-specific feature streams before their power and state heads. The architectural trade-off is therefore how to retain appliance evidence and relational context at modest computational cost.

Architecture alone does not determine joint learning. Gradient normalization balances task updates directly [25], whereas SAMNet uses a power–state loss ratio [17]. Background recomposition addresses a different issue: exposure to residual demand beyond the target set [16]. Together, these studies motivate evaluating architectural sharing, appliance-wise power–state balance, and residual-background coverage.

III. Methodology

A. Problem Formulation

Given aggregate power x_t, the task is to estimate the powers and ON–OFF states of N target appliances. The recorded aggregate also includes unmonitored demand b_t and measurement discrepancies ϵ_t, leaving the unobserved residual

$$x_t - \sum_{i=1}^{N} P_{t,i} = b_t + \epsilon_t. \tag{3}$$

For an aggregate window $\mathbf{x}$ of length T, the joint mapping in (2) produces paired outputs for every appliance and time step:

$$f_{\boldsymbol{\theta},t}(\mathbf{x}) = \underbrace{\big(\hat{y}_{t,1}, \hat{s}_{t,1}, \ldots, \hat{y}_{t,N}, \hat{s}_{t,N}\big)}_{N \text{ appliance power–state pairs}}, \quad t = 1, \ldots, T. \tag{4}$$

Here $\hat{y}_{t,i}$ estimates appliance power on the standardized scale, and $\hat{s}_{t,i} \in [0, 1]$ is its ON-state probability. Power estimation is a regression task, whereas state detection is multi-label classification because several appliances can be ON simultaneously. Binary ON–OFF labels summarize activity without resolving internal operating modes.

B. Model Architecture

Figure 3 presents MART-NILM, combining the private-stream interaction body in Fig. 2(c) with the state-gated heads in Fig. 1(b). The components are described below in processing order, using the same names as the architecture ablations in Table V.

Multi-scale CNN: Parallel convolutional branches [26] extract local features at complementary time scales from the standardized aggregate $\bar{\mathbf{x}}$:

$$\mathbf{Z} = \mathcal{F}([\mathcal{B}_3(\bar{\mathbf{x}}); \mathcal{B}_5(\bar{\mathbf{x}}); \mathcal{B}_9(\bar{\mathbf{x}})]) + \mathcal{S}(\bar{\mathbf{x}}). \tag{5}$$

Here $\mathcal{B}_k$ is a kernel-k convolution followed by instance–batch normalization (IBN) [27] and a rectified linear unit (ReLU); $[\cdot;\cdot]$ denotes channel concatenation. The 1×1 fusion $\mathcal{F}$ combines the branches, while the projected shortcut $\mathcal{S}$ retains a direct input path. IBN applies instance normalization (IN) and batch normalization (BN) to separate channel halves.

Feature encoder: Two convolution–IBN–ReLU stages combine the local features into a shared representation $\mathbf{H}^{(0)} \in \mathbb{R}^{128 \times T}$. Feature extraction is performed once for all appliances.

Temporal encoder: Residual dilated TCN blocks [28] extend temporal context:

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \mathcal{T}_{2^\ell}(\mathbf{H}^{(\ell)}), \qquad \ell \in \{0, 1, 2, 3, 4\}. \tag{6}$$

Here $\mathcal{T}_d$ comprises a dilation-d convolution, BN, ReLU, and dropout. Increasing dilation expands context, while residual connections and symmetric padding retain the incoming features and temporal alignment. The output is $\mathbf{H} = \mathbf{H}^{(5)}$.

Appliance decoders: Each decoder applies a channel gate, inspired by channel recalibration [29], and a local residual transformation to obtain an appliance-specific representation $\mathbf{F}_i$. These private streams adapt the shared temporal features to individual load characteristics.

Relation attention: The relation module draws on the cross-task information-sharing principle of PAD-Net [24] to exchange context among appliance-specific feature streams.

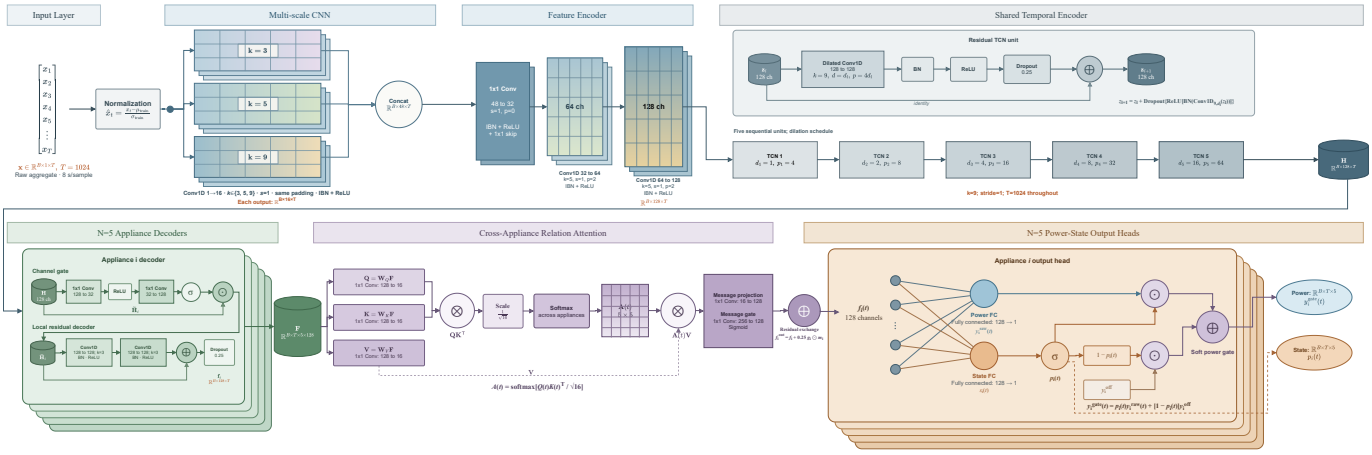


Fig. 3. Model Architecture of MART-NILM.

Scaled dot-product attention [30] is evaluated across appliances at each time t:

$$\mathbf{A}_t = \mathrm{softmax}\left(\frac{\mathbf{Q}_t\mathbf{K}_t^{\mathsf{T}}}{\sqrt{d}}\right), \qquad \mathbf{M}_t = \mathbf{A}_t\mathbf{V}_t. \tag{7}$$

Here $\mathbf{Q}_t$, $\mathbf{K}_t$, and $\mathbf{V}_t$ are learned projections of the stacked appliance representations, and d is their projection width. The attended context is projected back to the feature dimension and added to each private stream through a gated residual connection.

Joint power and state prediction: For each appliance, two time-shared fully connected (FC) heads predict raw standardized power $\hat{y}_i^{\mathrm{raw}}(t)$ and a state logit $a_i(t)$. Following the state-gating principle of the subtask-gated network (SGN) [22], the on-state probability $\hat{s}_i(t) = \mathrm{sigmoid}(a_i(t))$ softly gates the power prediction:

$$\hat{y}_i(t) = \hat{s}_i(t)\hat{y}_i^{\mathrm{raw}}(t) + [1 - \hat{s}_i(t)]y_i^{(0)}, \tag{8}$$

where $y_i^{(0)} = -\mu_i/\sigma_i$ is the standardized zero-power anchor. Hence, a low on-state probability smoothly moves the estimate towards zero power. Predictions are converted back to watts using the appliance training mean μ_i and standard deviation σ_i, and negative values are clipped to zero. The gate couples the state and power predictions while retaining a separate waveform regressor.

C. Loss Function

The model is trained jointly over all appliances, as in multi-task NILM [11], [13], [15]. Let $\mathcal{L}_{\mathrm{power},i}$ and $\mathcal{L}_{\mathrm{state},i}$ denote the power and state losses for appliance i, respectively. The joint objective is

$$\mathcal{L} = \sum_{i=1}^{N} \left(\mathcal{L}_{\mathrm{power},i} + \lambda r_i \mathcal{L}_{\mathrm{state},i}\right), \qquad \lambda = 0.8. \tag{9}$$

Here r_i is an appliance-wise balancing factor defined below.

Power loss: The power objective combines waveform fidelity, switching fidelity, and window-energy consistency:

$$\mathcal{L}_{\mathrm{power},i} = \underbrace{\mathcal{L}_{\mathrm{wave},i}}_{\text{power-level fidelity}} + \underbrace{0.15\,\mathcal{L}_{\mathrm{trans},i}}_{\text{transition fidelity}} + \underbrace{0.25\,\mathcal{L}_{\mathrm{RE},i}}_{\text{energy consistency}}. \tag{10}$$

The three terms respectively fit instantaneous power levels, changes around appliance activity, and the total energy within each window.

The waveform and transition terms both use MSE, applied to power values and temporal differences, respectively. Let $y_i(t)$ and $\hat{y}_i(t)$ denote standardized target and predicted power, and let $z_i(t) \in \{0, 1\}$ denote the on–off label. For a temporal index set $\mathcal{S}$, define

$$\mathrm{MSE}_{\mathcal{S}}(u, v) = \frac{1}{\max(|\mathcal{S}|, 1)} \sum_{t\in\mathcal{S}} \big(u(t) - v(t)\big)^2. \tag{11}$$

The waveform term evaluates power errors over all samples $\mathcal{T}_0 = \{1, \ldots, T\}$, on-state samples $\mathcal{T}_i^{\mathrm{on}} = \{t \in \mathcal{T}_0 : z_i(t) = 1\}$, and off-state samples $\mathcal{T}_i^{\mathrm{off}} = \{t \in \mathcal{T}_0 : z_i(t) = 0\}$:

$$\begin{aligned}\mathcal{L}_{\mathrm{wave},i} = {} & \mathrm{MSE}_{\mathcal{T}_0}(y_i, \hat{y}_i) + \mathrm{MSE}_{\mathcal{T}_i^{\mathrm{on}}}(y_i, \hat{y}_i) \\ & + 0.5\ \mathrm{MSE}_{\mathcal{T}_i^{\mathrm{off}}}(y_i, \hat{y}_i).\end{aligned} \tag{12}$$

The first term measures global waveform fidelity; the on-state term prevents sparse activations from being dominated by off samples, and the lower-weight off-state term constrains inactive predictions.

To additionally constrain changes in power, let $\Delta y_i(t) = y_i(t) - y_i(t-1)$ and define $\Delta\hat{y}_i(t)$ analogously. The transition term compares these differences whenever the appliance is on at either adjacent sample, over $\mathcal{T}_i^{\Delta} = \{t \in \{2, \ldots, T\} : z_i(t) = 1 \text{ or } z_i(t-1) = 1\}$:

$$\mathcal{L}_{\mathrm{trans},i} = \mathrm{MSE}_{\mathcal{T}_i^{\Delta}}(\Delta y_i, \Delta\hat{y}_i). \tag{13}$$

Finally, a stabilized relative energy term constrains the window total. With physical-unit powers $P_i(t) = [\sigma_i y_i(t) + \mu_i]_+$ and $\hat{P}_i(t) = [\sigma_i\hat{y}_i(t) + \mu_i]_+$, it is

$$\mathcal{L}_{\mathrm{RE},i} = \frac{\left|\sum_{t=1}^{T}(\hat{P}_i(t) - P_i(t))\right|}{\sum_{t=1}^{T} P_i(t) + TP_0}, \qquad P_0 = 10\ \mathrm{W}. \tag{14}$$

Here P_0 is a stabilizing power scale that limits the weight of near-zero-energy windows while retaining a penalty for false power in all-off windows. The constant sampling interval cancels in the ratio.

State loss: The state objective combines class-balanced state detection with explicit false-positive control:

$$\mathcal{L}_{\text{state},i} = \underbrace{\mathcal{L}_{\text{BCE},i}}_{\text{class-balanced state detection}} + \underbrace{\mathcal{L}_{\text{fp},i}}_{\text{false-positive control}}. \tag{15}$$

For a mini-batch of B windows, let $\hat{s}_{b,t,i} \in (0,1)$ be the predicted on-state probability and $z_{b,t,i} \in \{0,1\}$ the target label. The class-balanced detection term is a weighted binary cross-entropy, with $w_i = \min\{(1-\pi_i)/\pi_i, 12\}$ determined by the training on-state proportion π_i:

$$\mathcal{L}_{\text{BCE},i} = -\frac{1}{BT}\sum_{b,t}\Big[w_i z_{b,t,i}\log \hat{s}_{b,t,i} + (1-z_{b,t,i})\log(1-\hat{s}_{b,t,i})\Big]. \tag{16}$$

The false-positive term penalizes confident on-state predictions on off samples:

$$\mathcal{L}_{\text{fp},i} = \frac{\sum_{b,t}(1-z_{b,t,i})\hat{s}^2_{b,t,i}}{\max\left(\sum_{b,t}(1-z_{b,t,i}), 1\right)}. \tag{17}$$

Appliance-wise task balance: To account for the different numerical scales of regression and classification losses, a bounded loss-ratio weighting is used. It is motivated by loss-ratio balancing in SAMNet [17] and differs from gradient-norm balancing in GradNorm [25]. The global reference ratio g bounds the appliance-specific ratio r_i:

$$g = \frac{\sum_j \mathcal{L}_{\text{power},j}}{\sum_j \mathcal{L}_{\text{state},j}}, \qquad r_i = \min\left\{3g, \max\left\{\frac{g}{3}, \frac{\mathcal{L}_{\text{power},i}}{\mathcal{L}_{\text{state},i}}\right\}\right\}. \tag{18}$$

At each mini-batch update, g and r_i are evaluated from the current losses and held fixed during back-propagation. The three-fold bounds limit each local ratio to $[g/3, 3g]$, preventing extreme weights from dominating optimization.

D. Focal-Event Background-Stratified Mixing

Training uses recorded and recomposed aggregate windows with equal probability. Recorded windows preserve the observed joint demand. Recomposed windows pair appliance events with residual backgrounds from other training windows, reducing reliance on house-specific event–background correlations [6], [16]. For a recorded sequence, the non-target residual is

$$R_t = \left[x_t - \sum_{i=1}^{N} P_{t,i}\right]_+. \tag{19}$$

For each recomposed window, the residual and appliance traces are drawn from independently selected training windows:

$$\tilde{x}_t = R_t^{(k_0)} + \sum_{i=1}^{N} P_{t,i}^{(k_i)}, \qquad \tilde{P}_{t,i} = P_{t,i}^{(k_i)}, \quad \tilde{z}_{t,i} = z_{t,i}^{(k_i)}, \tag{20}$$

where $k_0, k_1, \ldots, k_N$ denote independent window indices. One focal appliance $i^\star$ is selected uniformly, and its trace is drawn from a window containing at least one on sample. The residual is sampled from a uniformly selected non-empty bin of median residual power; all remaining appliance traces are sampled independently. Thus, rare events occur against varied recorded backgrounds while their selected power and state labels are retained. Recomposition is used only during training; validation and test use recorded aggregates and aligned labels.

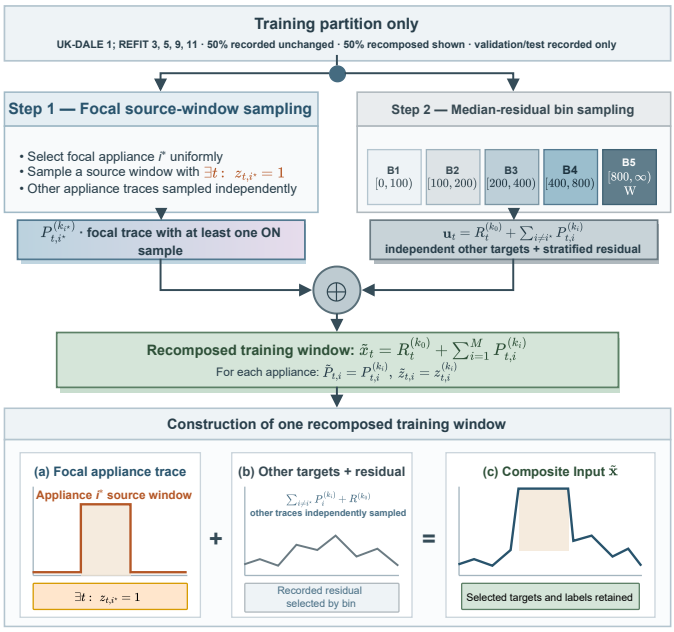


Fig. 4. Focal-event stratified mixing recomposes $\tilde{x}_t$ from an on-event focal trace, a power-stratified residual background, and independently sampled remaining appliance traces.

IV. Experimental Evaluation

A. Dataset

Two public datasets are used: UK-DALE [31] and REFIT [32]. The target appliances are the kettle, fridge, dishwasher, washing machine, and microwave, and all signals are resampled at 8 s.

The experiment uses a whole-house split: UK-DALE house 1 and REFIT houses 3, 5, 9, and 11 for training; UK-DALE house 5 and REFIT house 2 for validation; and UK-DALE house 2 and REFIT house 20 for testing. Each house contributes one continuous 5-week block. The held-out blocks are centred within the previously established eight-week recording intervals and are selected without reference to appliance events or model performance. Normalization statistics are estimated from the training houses only.

B. Data Preprocessing

After 8 s resampling, no additional temporal alignment is applied; residual timing discrepancies remain a measurement limitation. Rows missing any required aggregate, appliance-power, or state-label value are excluded, and windows do not cross timestamp discontinuities.

Let n index windows and t index samples within a window. The aggregate mean and standard deviation, μ_x and σ_x, and

TABLE I
ON–OFF LABEL PARAMETERS. THE FINAL COLUMN LISTS UK-DALE / REFIT VALUES.

Appliance	τ (W)	G (s)	D (s)
Kettle	200	0	12 / 12
Fridge†	50	12	60 / 60
Dishwasher†	50	1800	960 / 960
Washing machine†	20	960	960 / 960
Microwave	200	30	12 / 6

† For each sample, a five-sample local-median filter sets its label-generation power to zero when at least 60% of its four surrounding values are below 50 W and the sample exceeds both three times their median and 100 W. Endpoint windows use edge replication.

the corresponding appliance-i statistics, μ_i and σ_i, are estimated from the training houses. The input and target powers are standardized as

$$\bar{x}_{n,t} = \frac{x_{n,t} - \mu_x}{\sigma_x}, \qquad y_{n,t,i} = \frac{P_{n,t,i} - \mu_i}{\sigma_i}. \tag{21}$$

State labels are precomputed in the CSVs before windowing. For appliances marked in Table I, isolated spikes are first removed on a label-only copy of the resampled power trace. Each sample is then labeled on when this trace reaches its appliance-specific threshold τ_i. OFF gaps of at most G_i are merged; subsequently, complete ON runs shorter than D_i are removed. Runs touching a sequence boundary are retained because their full duration is unobserved. Thus, τ_i, G_i, and D_i denote the ON-power threshold, maximum OFF-gap duration, and minimum ON duration for appliance i, respectively. Table I specifies their physical-time values, which are converted to 8 s samples by half-up rounding before labeling. UK-DALE house 5 uses a 30 s fridge gap.

C. Experimental Setup

All experiments were conducted on a single NVIDIA GeForce RTX 4090 graphics processing unit (GPU; 24 GB) using PyTorch 2.6.0 with Compute Unified Device Architecture (CUDA) 12.4. MART-NILM is compared with UNet-NILM [11], MAT-Conv [15], and the multiappliance-task network of Sun *et al.* [14]. Each baseline retains the architecture, objective, and training hyperparameters of its source paper, while all models use the common house-held-out data protocol. Table II lists their settings. MART-NILM uses weight decay 10^{-4} and unit-norm gradient clipping. Its learning rate is reduced by a factor of 0.5 after 12 epochs without validation improvement, to a minimum of 10^{-5}. Training is limited to 150 epochs; early stopping is disabled before epoch 100 and thereafter uses a patience of 55 epochs. Its 1024-sample input and output windows each cover about 136.5 min.

For MART-NILM, checkpoint selection uses only the validation houses. Average precision (AP) is computed from the continuous state probabilities, and the retained checkpoint minimizes

$$S_{\mathrm{val}} = \mathrm{MAE}_{\mathrm{val}}^{\mathrm{norm}} + 1 - \mathrm{AP}_{\mathrm{val}}^{\mathrm{macro}}, \tag{22}$$

where $\mathrm{MAE}_{\mathrm{val}}^{\mathrm{norm}}$ is the macro mean absolute error over appliances in standardized power coordinates, and the AP term summarizes the precision–recall curve without selecting a decision threshold. For each model and appliance i, let $\hat{s}_{t,i} \in [0,1]$ denote the predicted ON probability at sample t. After checkpoint selection, an appliance-specific threshold $\tau_i \in \{0.05, 0.06, \ldots, 0.98\}$ is selected on the validation houses to maximize appliance-level F1 after the fixed temporal decoding rules are applied. At test time, the decoded state label is

$$\hat{z}_{t,i} = \mathbb{I}[\hat{s}_{t,i} \geq \tau_i], \tag{23}$$

where $\mathbb{I}[\cdot]$ is the indicator function. For the microwave, the fixed decoder removes predicted ON runs shorter than 32 s and merges OFF gaps of at most 24 s. The decoded state is also applied to the power output, $\hat{P}_{t,i}^{\mathrm{eval}} = \hat{z}_{t,i}\hat{P}_{t,i}$. Thresholds and temporal rules are frozen before testing and applied unchanged to both test houses; test labels are not used to select τ_i.

TABLE II
TRAINING WINDOWING AND BATCH SETTINGS. ALL WINDOW LENGTHS AND STRIDES ARE IN SAMPLES.

Model	Input	Output	Stride	Batch	Opt.	Max. epochs
UNet-NILM [11]	100	1	1	256	Adam	50
MAT-Conv [15]	864	64	64	64	Adam	200
Multi-task net. [14]	480	480	240	32	Adam	100
MART-NILM	1024	1024	512	64	Adam	150

D. Evaluation Metrics

Each appliance is evaluated on power and on the on–off state. For an evaluated record of length T_{rec}, P_t is the recorded power in watts and $\tilde{P}_t = \hat{P}_{t,i}^{\mathrm{eval}}$ is the state-decoded prediction. The label z_t equals 1 when the appliance is recorded on and 0 when it is off. $\hat{z}_t$ is the predicted label. The Average column in Table III is the mean of the five appliance scores.

Results are reported separately for UK-DALE house 2 and REFIT house 20. Within each house, the Avg. column is the macro average over the five appliances; architecture-ablation summaries give the two test houses equal weight. All reported results are mean $\pm$ standard deviation over three random seeds. These seeds quantify training variability on the fixed house split, rather than variability across households. Because the test houses were inspected during method development, they provide a development-informed cross-house benchmark rather than an untouched confirmatory evaluation [33]. The evaluation therefore considers complementary power-estimation and state-detection metrics.

1) Power: Power error is reported using mean absolute error (MAE), ON-state MAE (ON MAE), and blockwise signal aggregate error (SAE). MAE uses every sample; ON MAE is the same quantity evaluated only at samples with $z_t = 1$, so that reconstruction quality during appliance operation remains visible:

$$\mathrm{MAE} = \frac{1}{T_{\mathrm{rec}}} \sum_{t=1}^{T_{\mathrm{rec}}} \left| P_t - \tilde{P}_t \right|. \tag{24}$$

Following MATNilm [15], SAE measures the mean absolute error of average power over non-overlapping 160 min blocks ($L = 1200$ samples):

$$\mathrm{SAE} = \frac{1}{K}\sum_{k=1}^{K}\left|\frac{1}{L}\sum_{t=(k-1)L+1}^{kL}\left(P_t - \tilde{P}_t\right)\right|, \qquad K = \left\lfloor \frac{T_{\mathrm{rec}}}{L} \right\rfloor. \tag{25}$$

2) On-Off State: The recorded label z_t follows the rules in Table I, and $\hat{z}_t$ is the predicted binary state. For each appliance, TP (true positives) counts samples correctly identified as on ($z_t = 1, \hat{z}_t = 1$), FP (false positives) counts inactive samples falsely detected as on ($z_t = 0, \hat{z}_t = 1$), and FN (false negatives) counts active samples missed as off ($z_t = 1, \hat{z}_t = 0$):

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad \mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{26}$$

$$\mathrm{F1} = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{27}$$

Precision is the fraction of predicted on samples that are truly on, and recall is the fraction of actual on samples detected. F1 is their harmonic mean; true negatives (TN, $z_t = 0, \hat{z}_t = 0$) are excluded to prevent long idle background periods from artificially inflating the score. To evaluate state ranking independently of the calibrated decision threshold, we also report AP from the continuous state probability. Let $\mathcal{P}_n$ and $\mathcal{R}_n$ denote precision and recall after the nth distinct probability threshold, with $\mathcal{R}_0 = 0$. The non-interpolated AP is

$$\mathrm{AP} = \sum_{n} \left(\mathcal{R}_n - \mathcal{R}_{n-1}\right)\mathcal{P}_n. \tag{28}$$

Unlike F1, AP does not use the appliance-specific threshold or the temporal state postprocessing fitted on the validation houses.

V. Results

A. Power Disaggregation and State Detection

Table III compares each model on UK-DALE and REFIT. DW, WM, and MW are the dishwasher, washing machine, and microwave. Power errors are reported in watts. Each entry is the mean and standard deviation over three seeds; entries with ± 0.0, ± 0.00, or ± 0.000 indicate values below the reported display precision. When a seed produces no positive prediction, its precision is defined as zero. The scores are defined in Section IV-D. Lower power error is better. Higher AP, precision, recall, and F1 are better.

The threshold-independent results agree with the calibrated F1 comparison. MART-NILM obtains macro-AP values of 0.926 on UK-DALE and 0.762 on REFIT, compared with 0.812 and 0.576 for MAT-Conv, 0.753 and 0.474 for UNet-NILM, and 0.681 and 0.440 for the multiappliance-task network, respectively. It also gives the highest reported AP for every appliance, with a tie at the reported precision for the UK-DALE microwave. The state-detection advantage therefore persists across the full precision–recall ranking and cannot be attributed only to the calibrated decision thresholds.

Figure 5 compares the four models on identical selected intervals. MART-NILM tracks the activation plateaus and switching patterns in these examples, although amplitude errors remain visible. The REFIT fridge and washing-machine panels are selected success cases; overall performance is assessed over the complete test recordings in Table III.

B. Computational Cost

Table IV reports the number of trainable parameters, checkpoint size, and training time of each model on one GPU. Checkpoint size is measured from the saved `best.pt` file using $1\,\mathrm{MB} = 1024^2$ bytes.

Under the common implementation and hardware protocol, MART-NILM contains 82.5% fewer trainable parameters than the multiappliance-task network and reduces mean training time from 16493.8 s to 485.2 s. This comparison concerns computational cost; Table III separately reports predictive performance.

C. Ablation Study

Architecture and training effects are evaluated separately. Every architecture variant in Table V remains one joint five-appliance network and retains five power and five state outputs. A removed module is replaced only by an identity mapping or the pointwise projection required to preserve tensor dimensions. Each row therefore removes or simplifies one component shown in Fig. 3, while all other settings remain fixed. Checkpoints are selected on the validation houses. The table then reports frozen evaluation on UK-DALE house 2 and REFIT house 20, with the two house-level metrics averaged equally.

The multi-scale CNN is reduced to its central kernel while preserving its width and residual path. Removing the feature encoder retains only a pointwise channel projection, and removing the temporal encoder uses an identity mapping. The appliance-decoder ablation sends the shared temporal representation directly to five independent power/state readouts. The final variant retains these appliance-specific readouts but removes information exchange between them. These replacements preserve the joint prediction task and isolate the contribution of each top-level module.

The shared temporal encoder gives the largest contribution: its removal increases MAE by 1.44 W and ON MAE by 45.13 W, while macro-F1 falls from 0.796 to 0.710. Removing the appliance-specific decoders increases ON MAE by 22.90 W and reduces macro-F1 by 0.013. The feature encoder provides a smaller but consistent benefit to ON-state reconstruction and detection. The multi-scale branches have little effect on the aggregate averages, but their benefit is concentrated on the sparse microwave task: on validation UK-DALE 5 the single-scale variant increases microwave ON MAE from 589.49 to 696.45 W, and on REFIT 20 microwave F1 falls from 0.549 to 0.512. Relation attention also exhibits a task trade-off. Removing it lowers average ON MAE, but reduces macro-F1 and lowers REFIT 20 microwave F1 from 0.549 to 0.496. Waveform inspection agrees with the numerical results: removing temporal encoding produces prolonged false fridge activations and poorly aligned microwave plateaus,

TABLE III

POWER AND ON–OFF RESULTS (MEAN ± STANDARD DEVIATION OVER THREE SEEDS). BOLD DENOTES THE BEST DISPLAYED MEAN, INCLUDING TIES.

Metric	Method	UK-DALE						REFIT					
		Kettle	Fridge	DW	WM	MW	Avg.	Kettle	Fridge	DW	WM	MW	Avg.
MAE	UNet-NILM	9.2±2.1	29.1±0.9	22.6±5.1	9.1±0.1	6.1±0.3	15.2±1.1	8.6±0.9	35.0±1.7	12.1±1.1	7.6±0.1	7.4±0.3	14.2±0.3
	MAT-Conv	5.5±1.6	**18.4±0.7**	8.9±2.1	9.6±1.4	5.3±0.4	9.5±1.1	**5.4±0.4**	40.8±0.1	5.4±0.8	6.2±2.3	**6.0±0.7**	12.8±0.6
	Multi-task net.	8.8±1.7	34.4±3.0	14.7±6.2	15.5±5.0	6.3±0.1	15.9±1.3	10.9±1.6	35.6±1.8	9.0±1.2	18.0±7.3	6.7±0.1	16.1±0.8
	MART-NILM	**3.8±0.3**	20.3±0.4	**4.4±0.4**	**5.3±0.3**	**3.4±0.1**	**7.5±0.1**	5.7±0.1	**32.6±1.5**	**4.8±0.3**	**3.6±0.2**	7.3±0.4	**10.8±0.4**
SAE	UNet-NILM	8.4±2.6	24.5±2.5	21.1±6.5	9.0±0.1	6.0±0.4	13.8±1.3	5.6±1.1	25.9±2.6	9.8±0.3	7.1±0.3	7.0±0.2	11.1±0.7
	MAT-Conv	4.3±1.2	**10.0±1.0**	6.7±1.4	8.8±1.3	5.0±0.4	6.9±0.8	2.9±0.3	23.7±0.6	4.5±1.0	5.0±1.8	**5.2±0.5**	8.3±0.5
	Multi-task net.	7.3±1.4	32.1±3.5	12.0±5.9	14.5±4.9	6.2±0.1	14.4±1.4	7.1±1.1	21.7±4.4	7.9±1.3	13.0±5.5	6.6±0.0	11.3±0.7
	MART-NILM	**2.9±0.2**	13.6±0.6	**2.8±0.5**	**4.1±0.5**	**3.0±0.1**	**5.3±0.1**	**2.3±0.2**	**15.0±1.6**	**3.6±0.1**	**2.9±0.2**	5.6±0.3	**5.9±0.4**
ON MAE	UNet-NILM	1296.5±443.8	48.4±4.0	747.7±180.9	462.6±7.3	1178.8±93.3	746.8±110.0	1289.1±190.7	63.4±3.4	398.3±93.1	306.1±6.7	1196.7±58.0	650.7±59.1
	MAT-Conv	547.5±180.2	**22.1±1.6**	206.9±46.2	385.7±59.1	946.4±99.8	421.7±75.2	850.7±44.4	46.3±2.5	208.2±21.0	195.6±10.1	1062.2±90.6	472.6±29.3
	Multi-task net.	807.3±69.1	61.0±7.1	449.8±212.8	478.2±28.0	1215.2±19.2	602.3±44.3	1012.0±65.7	59.1±6.4	437.1±91.3	306.7±17.6	1224.2±5.6	607.8±16.5
	MART-NILM	**388.0±38.1**	26.8±1.0	**96.0±9.6**	**164.0±19.2**	**208.5±21.1**	**176.7±7.7**	**791.8±22.3**	**40.7±1.9**	**156.9±17.9**	**167.7±2.2**	**388.9±22.6**	**309.2±4.6**
AP	UNet-NILM	0.908±0.016	0.873±0.041	0.580±0.274	0.866±0.026	0.535±0.024	0.753±0.049	0.653±0.036	0.657±0.030	0.269±0.056	0.654±0.129	0.136±0.046	0.474±0.024
	MAT-Conv	0.968±0.020	0.951±0.006	0.868±0.039	0.555±0.191	**0.718±0.063**	0.812±0.051	0.801±0.018	0.537±0.005	0.736±0.040	0.653±0.066	0.152±0.046	0.576±0.024
	Multi-task net.	0.930±0.016	0.915±0.030	0.838±0.094	0.207±0.128	0.514±0.194	0.681±0.075	0.660±0.090	0.641±0.053	0.615±0.040	0.123±0.019	0.160±0.069	0.440±0.000
	MART-NILM	**0.997±0.001**	**0.969±0.001**	**0.963±0.002**	**0.985±0.008**	**0.718±0.033**	**0.926±0.008**	**0.848±0.008**	**0.737±0.025**	**0.766±0.009**	**0.904±0.033**	**0.557±0.045**	**0.762±0.018**
Precision	UNet-NILM	0.80±0.09	0.74±0.09	0.55±0.19	0.84±0.09	0.57±0.07	0.70±0.03	0.63±0.08	0.60±0.12	0.26±0.06	**0.89±0.05**	0.18±0.07	0.51±0.04
	MAT-Conv	0.82±0.10	0.76±0.03	0.47±0.05	0.61±0.18	**0.71±0.09**	0.67±0.08	0.70±0.10	0.59±0.00	0.42±0.06	0.59±0.07	0.16±0.03	0.49±0.04
	Multi-task net.	0.74±0.08	0.87±0.01	0.51±0.16	0.22±0.09	0.55±0.49	0.58±0.11	0.57±0.08	0.63±0.00	0.54±0.09	0.14±0.02	0.37±0.28	0.45±0.04
	MART-NILM	**0.94±0.02**	**0.88±0.01**	**0.63±0.06**	**0.92±0.02**	**0.71±0.00**	**0.82±0.01**	**0.76±0.01**	**0.64±0.00**	**0.62±0.04**	0.85±0.07	**0.43±0.03**	**0.66±0.01**
Recall	UNet-NILM	0.87±0.08	0.82±0.14	0.65±0.21	0.78±0.04	0.57±0.07	0.74±0.05	0.73±0.03	0.65±0.18	0.46±0.07	0.43±0.13	0.37±0.04	0.53±0.07
	MAT-Conv	0.97±0.01	**0.98±0.02**	0.98±0.02	0.49±0.18	0.67±0.13	0.82±0.02	0.78±0.03	**0.93±0.06**	**0.83±0.02**	0.76±0.21	0.42±0.25	0.74±0.11
	Multi-task net.	0.92±0.03	0.94±0.01	0.89±0.14	0.56±0.06	0.08±0.07	0.68±0.05	0.75±0.05	0.78±0.01	0.77±0.08	0.50±0.12	0.03±0.02	0.57±0.01
	MART-NILM	**0.99±0.00**	0.94±0.01	**0.99±0.01**	**0.97±0.02**	**0.91±0.03**	**0.96±0.00**	**0.81±0.01**	0.79±0.02	0.77±0.04	**0.85±0.09**	**0.76±0.03**	**0.80±0.03**
F1	UNet-NILM	0.83±0.03	0.77±0.07	0.60±0.20	0.81±0.03	0.57±0.04	0.71±0.03	0.67±0.03	0.61±0.08	0.33±0.07	0.58±0.12	0.23±0.05	0.48±0.04
	MAT-Conv	0.89±0.06	0.86±0.01	0.63±0.04	0.54±0.17	0.68±0.06	0.72±0.05	0.73±0.05	**0.72±0.02**	0.55±0.05	0.65±0.06	0.23±0.08	0.58±0.02
	Multi-task net.	0.82±0.04	**0.91±0.00**	0.64±0.14	0.30±0.09	0.15±0.13	0.56±0.05	0.64±0.03	0.70±0.00	0.63±0.05	0.21±0.01	0.06±0.04	0.45±0.00
	MART-NILM	**0.96±0.01**	**0.91±0.00**	**0.77±0.04**	**0.94±0.02**	**0.80±0.01**	**0.88±0.01**	**0.79±0.00**	0.71±0.01	**0.69±0.01**	**0.85±0.05**	**0.55±0.03**	**0.72±0.02**

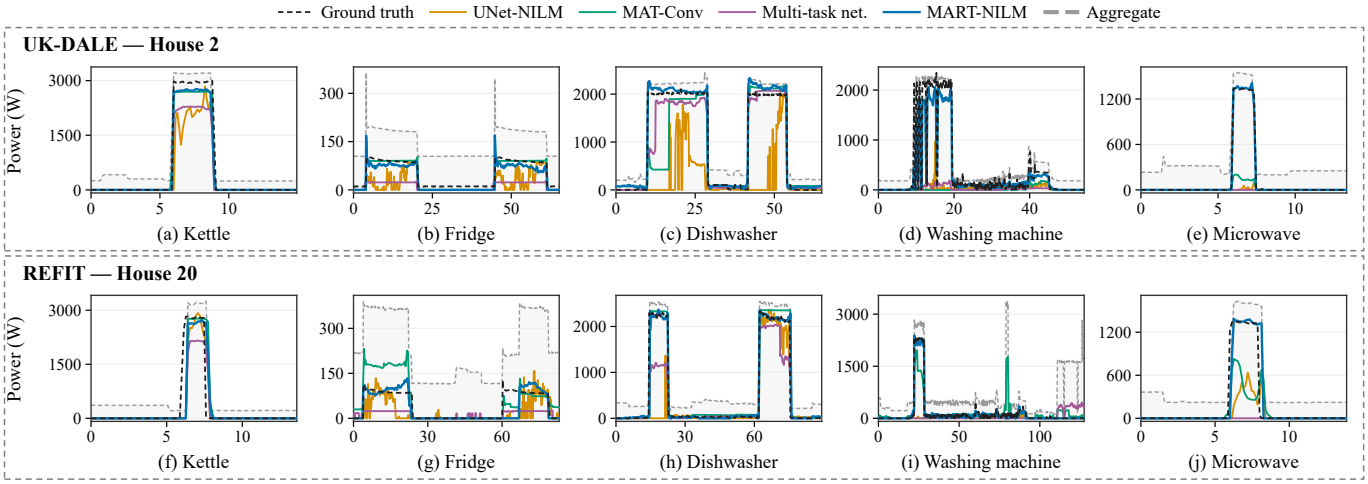


Fig. 5. Waveform comparisons on the two test houses.

TABLE IV

COMPUTATIONAL COST.

Model	Params (M)	Model size (MB)	Time (s)
UNet-NILM	5.832	22.299	17455.8
MAT-Conv	3.022	11.611	10354.9
Multi-task net.	7.829	59.793	16493.8
MART-NILM	**1.373**	**5.335**	**485.2**

while removing the appliance decoders increases fragmentation and amplitude error.

Table VI freezes the complete architecture and examines the two proposed training strategies. Standard joint training uses recorded windows and a fixed power–state weight. The next two rows add focal-event mixing and appliance-wise balancing independently, and complete MART-NILM combines both strategies.

TABLE V

ARCHITECTURE ABLATION ON THE TWO TEST HOUSES (THREE-SEED MEAN ± STANDARD DEVIATION).

Variant	Params (M)	MAE (W) ↓	ON MAE (W) ↓	SAE (W) ↓	Macro-F1 ↑
Single-scale CNN	1.372	9.22 ±0.18	241.75 ±8.76	5.66 ±0.17	0.795 ±0.001
w/o feature encoder	1.326	9.16 ±0.06	252.01 ±2.21	5.59 ±0.07	0.790 ±0.006
w/o temporal encoder	0.634	10.56 ±0.33	288.07 ±10.25	6.56 ±0.16	0.710 ±0.013
w/o appliance decoders	0.836	9.22 ±0.66	265.84 ±4.30	5.63 ±0.63	0.783 ±0.015
w/o relation attention	1.332	9.18 ±0.09	**230.84 ±14.69**	**5.55 ±0.04**	0.790 ±0.015
MART-NILM	1.373	**9.12 ±0.14**	242.94 ±5.35	5.58 ±0.17	**0.796 ±0.008**

Table VI supports both strategies, but also exposes their different effects. Focal-event mixing improves macro-F1 by

TABLE VI
TRAINING-STRATEGY ABLATION ON THE VALIDATION HOUSES (THREE-SEED MEAN ± STANDARD DEVIATION). † DENOTES A STRATEGY PROPOSED IN THIS WORK.

Training strategy	MAE (W) ↓	ON MAE (W) ↓	SAE (W) ↓	Macro-F1 ↑
Standard joint training	21.164 ±0.394	357.808 ±19.280	15.938 ±0.348	0.605 ±0.012
+ Focal-event mixing†	21.679 ±0.144	**301.623 ±16.196**	15.601 ±0.365	0.649 ±0.016
+ Appliance-wise bal.†	20.929 ±0.065	334.862 ±10.888	15.175 ±0.322	0.661 ±0.014
Complete MART-NILM†	**20.791 ±0.175**	315.408 ±10.405	**15.016 ±0.385**	**0.687 ±0.003**

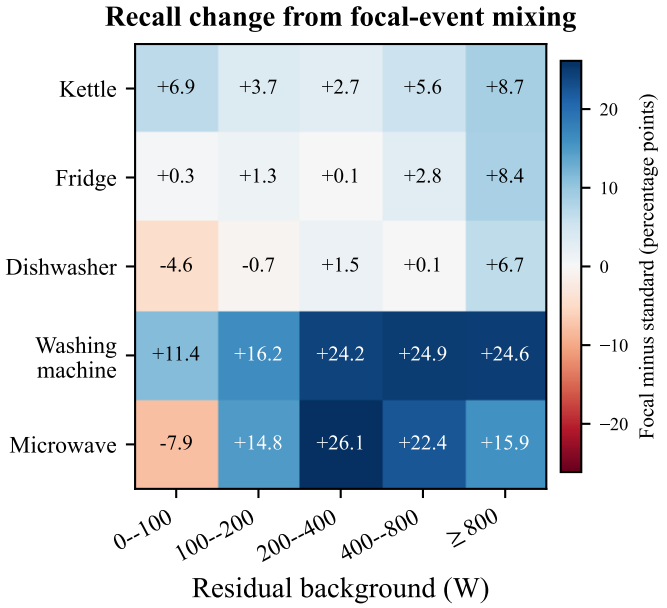


Fig. 6. Validation recall change from focal-event mixing across residual backgrounds (positive values denote improvement).

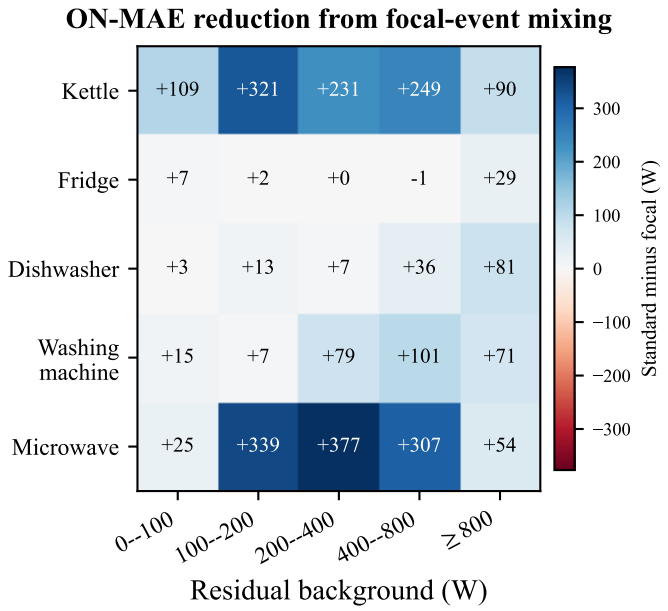


Fig. 7. Validation ON-MAE reduction from focal-event mixing across residual backgrounds (positive values denote lower error).

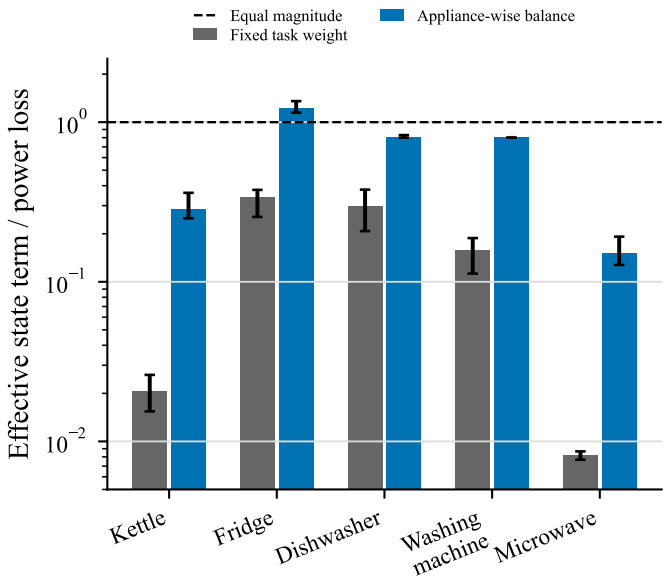


Fig. 8. Power–state scale alignment produced by appliance-wise balancing (median and interquartile range after epoch 20).

0.044 and ON MAE by 56.19 W relative to standard training, while its aggregate MAE is 0.52 W higher. It should therefore be interpreted as a coverage intervention rather than a generic regularizer. Appliance-wise balancing improves all four reported measures when used alone. Their combination gives the best MAE, SAE, and macro-F1, raising macro-F1 by 0.082 over standard joint training.

Figure 6 shows that focal-event mixing primarily improves event detection under non-trivial residual load. The largest gains occur for washing-machine and microwave events above 200 W. The effect is not universal: microwave recall decreases by 7.9 percentage points below 100 W, consistent with the unchanged overall microwave F1.

Figure 7 separates this detection effect from power estimation. Focal-event mixing substantially reduces ON MAE for kettle and microwave across several background intervals, and also improves the high-background estimates of dishwasher and washing machine. This explains why Table VI records better ON MAE despite a modest increase in aggregate MAE.

Figure 8 shows the distinct role of appliance-wise balancing. With a fixed task weight, the median effective state term is only 2.0% and 0.8% of the corresponding power loss for kettle and microwave. Balancing raises these ratios to 28.5% and 15.1%, respectively, while moving dishwasher and washing-machine state terms toward the power-loss scale. It therefore corrects an appliance-dependent task-scale mismatch rather than changing the event distribution.

The two strategies therefore address complementary limitations: event coverage and task-scale mismatch. Their combination raises macro-F1 by 0.082 over standard joint training. Microwave F1 changes from 0.352 ± 0.026 to 0.348 ± 0.027, so no microwave detection improvement is claimed.

VI. Conclusion

This paper presented MART-NILM for joint appliance power and state estimation from low-frequency aggregate measurements. Under the common five-week house-held-out benchmark, it achieves the lowest appliance-average power error and highest macro-F1 and average precision on both test houses among the reproduced models, using 1.373 million parameters. The shared temporal encoder and appliance-specific decoders provide the principal architectural gains. Focal-event mixing improves event coverage, while bounded appliance-wise balancing addresses unequal power–state loss scales; their combination improves validation detection without additional inference computation. The ablations also expose trade-offs, including limited microwave detection gains from the training strategies. These findings support compact joint estimation as a basis for residential load observability. Evidence remains restricted to two test houses inspected during development; broader household generalisation and downstream energy-management benefits require independent evaluation.

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